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## **Experiment No. 5: Demonstration of Hub Hardware and Setting up a Basic Local Area Network (LAN)**

### **Aim**

To identify the hardware interfaces of a Network Hub and to design, configure, and verify a basic Local Area Network (LAN) using a Star Topology to observe broadcast behavior in Cisco Packet Tracer.

### **Theory**

#### **1. Hub**

A **hub** is a basic networking device used to connect multiple computers or devices in a network. It operates at the **physical layer (Layer 1)** of the OSI model.

A hub does not store or filter data. It simply receives data from one device and **broadcasts it to all other connected devices**.

#### **Features:**

- Works as a central connection point
- No intelligence (does not check destination address)
- Data is sent to all ports

#### **2. Working Principle of a Hub**

The working of a hub is very simple:

- When a device sends data to the hub, the hub receives the signal
- The hub **does not analyze or modify the data**
- It simply **broadcasts the data to all connected devices**
- Each device checks whether the data is intended for it
- Only the correct device accepts the data; others ignore it

#### **Flow:**

Sender → Hub → All Devices in Network

#### **Conclusion:**

A hub works on a **broadcast mechanism**, which can lead to network congestion.

### 3. MAC Address

A **MAC (Media Access Control) address** is a unique physical address assigned to a network interface card (NIC) of a device.

- It is used for **identification of devices within a local network**
- It is assigned by the manufacturer

**Format:**

- 48-bit address
- Written in hexadecimal form

**Example:**

00:1A:2B:3C:4D:5E

**Conclusion:**

MAC address is a **hardware address** used for communication within a LAN.

### 4. IP Address

An **IP (Internet Protocol) address** is a unique numerical identifier assigned to each device connected to a network.

- It is used to identify and locate devices on a network

**Types:**

- **IPv4:** 192.168.1.1
- **IPv6:** 2001:0db8:85a3::8a2e:0370:7334

**Conclusion:**

IP address is a **logical address** used for communication over networks.

### 5. Local Area Network (LAN)

A **Local Area Network (LAN)** is a network that connects computers and devices within a **limited geographical area**, such as a home, school, or office.

**Features:**

- High data transfer speed
- Covers small area
- Easy to maintain

- Uses wired (Ethernet) or wireless (Wi-Fi) connections

**Examples:**

- Computer lab network
- Office network

## **6. Star Topology**

A **star topology** is a network structure in which all devices are connected to a **central device** such as a hub or switch.

**Working:**

- Each device communicates through the central device
- Data from one device goes to the central device and then to the destination

**Advantages:**

- Easy to install and manage
- Failure of one device does not affect others

**Disadvantages:**

- Failure of central device affects the entire network
- Requires more cable

**Conclusion:**

Star topology is widely used due to its simplicity and reliability.

## **Hardware Components Required**

- **PC-PT (x3):** End devices acting as network nodes.
- **Hub-PT:** A generic Physical Layer device used to interconnect the nodes.
- **Copper Straight-Through Cables:** The transmission media used for PC-to-Hub connectivity.

## **Hardware and Connection Setup**

- **Network ID:** 192.168.1.0
- **Subnet Mask:** 255.255.255.0
- **PC Naming Convention:**

- PC 1: **Batch\_Roll\_A** (e.g., S11\_01\_A)
- PC 2: **Batch\_Roll\_B** (e.g., S11\_01\_B)
- PC 3: **Batch\_Roll\_C** (e.g., S11\_01\_C)

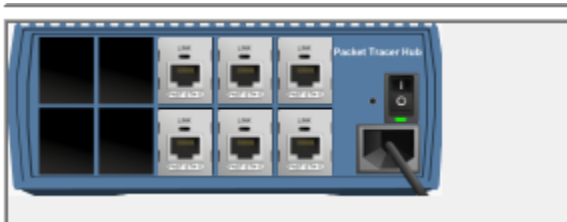
## Procedure

### Step 1: Physical Topology Design

1. Open Cisco Packet Tracer and select **Network Devices > Hubs > Hub-PT**. Place it in the center.
2. Select **End Devices > PC** and place four units around the Hub.
3. **Naming:** Click on each PC label and rename them following the convention: S11\_01\_A, S11\_01\_B, S11\_01\_C and S11\_01\_D.

### Step 2: Hardware Identification

1. Click on the **Hub** and select the **Physical** tab.
2. Observe the ports. Note that unlike the Switch, a Hub usually has fewer ports and no "intelligence" indicators (like console ports).



### Step 3: Physical Cabling

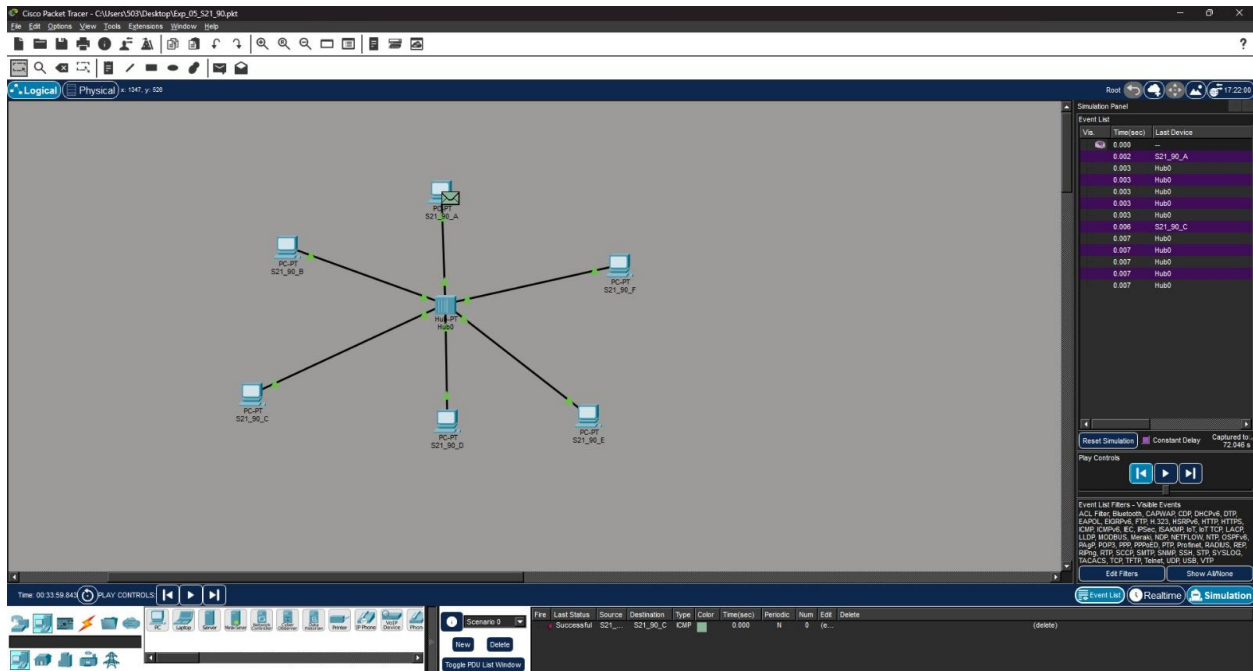
1. Select the **Copper Straight-Through** cable (solid black line).
2. Connect **S11\_01\_A (FastEthernet 0)** to **Hub (Port 0)**.
3. Repeat for the other three PCs using Hub ports **1, 2** and **3**.
4. **Observation:** Notice that the link lights turn **Green immediately**. Unlike a Switch, a Hub does not run STP (Spanning Tree Protocol), so there is no "Orange" waiting period.

### Step 4: Logical Addressing (IP Configuration)

1. Click **S11\_01\_A > Desktop > IP Configuration**.
2. Enter Static IP: 192.168.1.1 | Subnet: 255.255.255.0.
3. Configure **S11\_01\_B** with IP 192.168.1.2, **S11\_01\_C** with IP 192.168.1.3 and **S11\_01\_D** with IP 192.168.1.4.

### Step 5: Testing and Simulation (Crucial Step)

1. **Real-time:** Open the Command Prompt on **S11\_01\_A** and type ping 192.168.1.3.
2. **Simulation Mode:**
  - Click the **Simulation** tab and filter for **ICMP**.
  - Use the **Add Simple PDU (Envelope)** to click **S11\_01\_A** then **S11\_01\_C**.
  - Click **Play** and observe carefully: The packet goes to the Hub, and the Hub sends it to **all** PC B, PC C and PC D.
  - Observe the **Red Cross** on PC B and PC D (because they reject a packet not meant for them) and the **Green Check** on PC C.



## 6. Observations

- **Instant Connectivity:** Hub ports do not have a "learning" phase; they transition to green immediately upon connection.
- **Broadcast Behavior:** During simulation, the Hub was observed to forward the packet to every connected device. This wastes bandwidth and creates security risks.
- **Device Rejection:** End devices (PCs) are responsible for checking the destination MAC address. If it doesn't match, the PC discards the packet (indicated by the red cross in simulation).

## Conclusion

The experiment successfully demonstrates the hardware and functional limitations of a Hub. It concludes that a **Hub** is a non-intelligent Layer 1 device that relies on broadcasting, making it less efficient than a Switch for modern Local Area Networks.